Surgical Helmet

Mentors: Constantin Kleinbeck (constantin.kleinbeck@tum.de), Sasan, Veronica
Problem: Conventional surgical guidance relies on external screens, diverting attention from the task. Alternative, intuitive feedback methods could improve instrument alignment.
Background and Motivation: Helmet-integrated LEDs and auditory feedback offer a lightweight approach to spatial guidance without requiring a conventional AR display.
Goal: Develop a helmet-based guidance system for aligning a tracked probe with a target pose.
Proposed Solution:
Use OptiTrack to track probe pose and provide four feedback conditions:

Baseline: On-screen guidance, **Visual:** Helmet-integrated LEDs,
Auditory: Sonification of alignment errors, **Audiovisual:** Combined visual and auditory cues

Objectives:

- Integrate LEDs, speakers, and a microcontroller into existing surgical helmet.
- Design and test intuitive visual and auditory cues.
- Provide insights into alignment accuracy, completion time, and usability.

Tasks

- Design and build the helmet interface.
- Implement visual and auditory feedback.
- Test the system, conduct a user study and analyze results.
- Document findings and present the prototype.





1. AmbiGlasses



2. Stryker Surgical Helmet



3. 1-D Eyewear Prototype

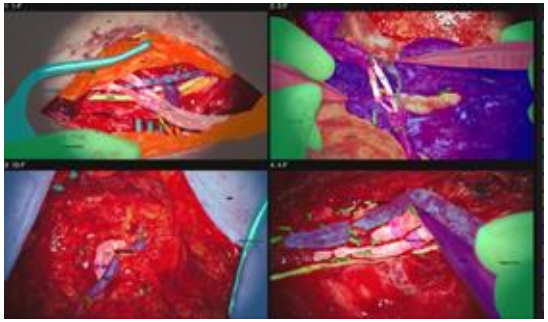
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Human-Centered Computing and Extended Reality

Leveraging Routinely Post-Anastomosis Videos for Data-Efficient Microsurgical Anatomy Segmentation

- **Mentors:** Luisa (luisa.theelke@tum.de), Sinan (LMU) , Irene (LMU)
- **Problem:** Annotated microsurgical video data are scarce, and segmenting vessels, nerves, and instruments is challenging in complex intraoperative scenes.
- **Background and Motivation:** Routine post-anastomosis microscope videos provide controlled views of the surgical scene and are already densely annotated by medical experts. This project investigates whether these recordings can serve as an efficient source of supervision for more complex full-surgery scenes.
- **Goal:** Develop and evaluate segmentation approaches that learn from annotated post-anastomosis videos and generalize to full microsurgical procedures
- **Proposed Solution:** Adapt pretrained segmentation foundation models using annotated documentation videos and evaluate generalization to full-surgery scenes, including domain-adaptation strategies.
- **Tasks and Objectives:**
    - Investigate and compare suitable approaches.
    - Quantify the domain gap between documentation and full-surgery videos.
    - Explore domain-generalization/ domain-adaptation strategies
    - Evaluate data efficiency and the benefit of limited target-domain supervision.
    - Analyze failure cases and visualize predictions as overlays on the original videos.



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Yuan, C., Jiang, J., Yang, K., Wu, L., Wang, R., Meng, Z., ... & Ban, Y. (2026). Systematic evaluation and guidelines for segment anything model in surgical video analysis. npj Digital Surgery, 1(1), 2.Tang, M., et al. (2024). Assessment of changes in vessel area during needle manipulation in microvascular anastomosis using a deep learning-based semantic segmentation algorithm. Neurosurgical Review, 47, 200.
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Human-Centered Computing and Extended Reality

World Models for Movement Prediction in the OR

Mentors: Hannah Schieber (hannah.schieber@tum.de), Benjamin Killeen

Problem: Existing surgical video models focus on recognition or scene-level prediction. The goal is to learn structured OR representations from single- or multi-camera videos and predict future surgical activities and room dynamics.

Background and Motivation: World models learn environmental dynamics to predict future states and simulate possible outcomes.

Goal: Develop a world model that learns a structured OR representations outside of endoscopic videos, predicts future scene states, and generalizes to unseen surgical scenarios while maintaining semantic and visual consistency.

Proposed Solution: Encode surgical videos into latent OR states and predict future states using learned dynamics, with surgical phase labels as semantic supervision. Train and evaluate on available datasets, measuring phase consistency, temporal coherence, visual quality, and generalization.

Objectives:

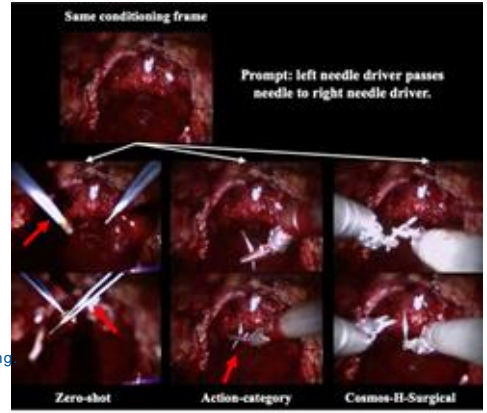
- Learn predictive OR scene representations using surgical video and phase-label supervision.
- Evaluate future-state predictions for semantic consistency, temporal coherence, and visual quality on unseen sequences.

Tasks

- *Data Preparation*: Prepare surgical data for model training and evaluation.
- World-State Representation: Design a latent representation of OR scenes.
- *Model Development*: Implement and train a world model to predict future latent states.
- *Future-State Generation:* Generate plausible future OR scenes from observed video clips.
- *Evaluation*: Assess prediction accuracy, consistency, temporal coherence, and generalization.







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Özsoy, E., Mamur, A., Tristram, F., Pellegrini, C., Wysocki, M., Busam, B., & Navab, N. (2026). Egoexor: An ego-exo-centric operating room dataset for surgical activity understanding. *Neurips*
Zhu, H., Liu, H., Zhao, Y., Ye, T., Chen, J., Yu, J., ... & Xie, E. (2026). Sana-wm: Efficient minute-scale world modeling with hybrid linear diffusion transformer. *arXiv*
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Human-Centered Computing and Extended Reality

Multimodal Force Feedback for Teleoperated Bone Cutting: Which Sensory Channels Work Best?

Mentor: Dimash
dimash.mukashev@tum.de

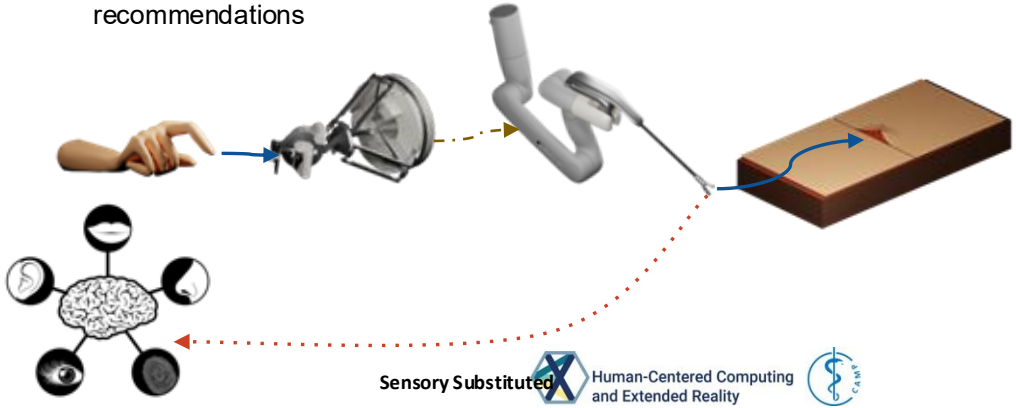


- **Topic Area:** Human-in-the-loop teleoperation and haptics for surgical robotics (embodiment, sensory substitution, mock-OR user studies, ROS-based testbeds)
- **Problem:** In teleoperated orthopedic surgery, the surgeon is physically separated from the patient. Simply mirroring the large, high-frequency tool–bone forces to the control hand overloads the operator, destabilizes the control loop, and masks the robot's guidance cues (sensory ambiguity).
- **Background and Motivation:** Robotic systems like Mako improve cutting precision but keep the surgeon at the bedside, hands-on. Moving to teleoperation requires rerouting interaction forces to other sensory channels (visual, auditory, tactile) — but each single channel has known limits (visual overload, auditory masking, low tactile bandwidth). Which combinations work, and for which task phases, is an open question.
- **Goal:** Systematically compare feedback modality combinations (visual / auditory / vibrotactile / kinesthetic) in a teleoperated cutting task, measuring task performance and cognitive load.

- **Proposed Solution:** Extend an existing ROS-based teleoperation testbed with a multimodal feedback pipeline; run a within-subject user study on standardized silicone/bone phantoms;

Tasks:

- Familiarize with the teleoperation testbed (Franka, Sigma.7, ROS)
- Implement 2–3 feedback mappings (e.g., sonification, vibrotactile display, visual overlay)
- Design and fabricate standardized phantom task (with lab support)
- Conduct a pilot user study (N≈12–15) comparing feedback conditions
- Analyze performance, force consistency, and workload; report design recommendations

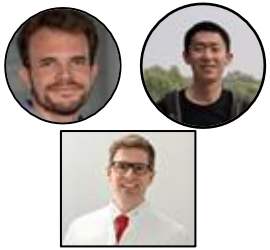


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Saracino, Arianna, et al. "Haptic feedback in the da Vinci Research Kit (dVRK): A user study based on grasping, palpation, and incision tasks." The International Journal of Medical Robotics and Computer Assisted Surgery 15.4 (2019): e1999.

Surgical Tattoos for Knee Arthroplasty

- **Topic Area:** Computer vision, fiducial markers
- **Problem:** Robotic-assisted knee surgeries (e.g., MAKO) are more precise than manual surgeries, but require dynamic tracking of the leg bones via optical markers which are pinned into the bones. This is necessary for the robotic assistance, but harms the bone.
- **Background and Motivation:** Less invasive would be visual markers that are applied to the skin, such as tattoos. However, soft tissue displacement occurs here, and the positions of the marker relative to the bone is not rigid. This makes it tricky to estimate the bone axes.
- **Goal:** Experimental evaluation of a non-invasive skin-based tracking method for tibia and femur axes
- **Proposed Solution:** Explore and compare skin-based marker approaches to ground truth data (e.g., using OptiTrack)
- **Tasks and Objectives:** Using an artificial knee, create a prototype which compares skin-based marker precision to ground truth femur and tibia axes.

- **Mentors**: Julian, Shiyu, Dr. med Dirk Müller



julian.kreimeier@tum.de
shiyu.li@tum.de





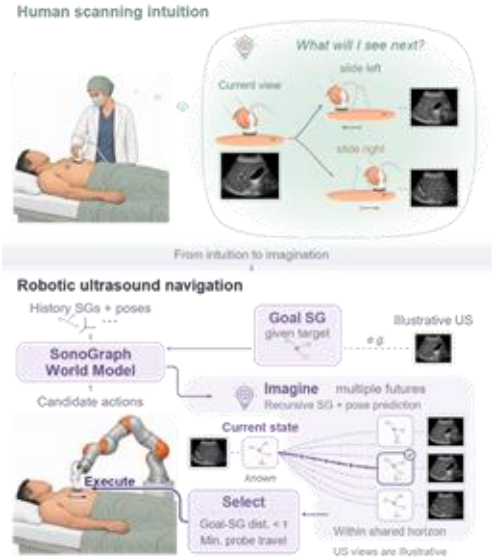


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Tsukada, S., Ogawa, H., Nishino, M., Kurosaka, K., & Hirasawa, N. (2019). Augmented reality-based navigation system applied to tibial bone resection in total knee arthroplasty. Journal of experimental orthopaedics, 6(1), 44
Tsukada, S., Ogawa, H., Nishino, M., Kurosaka, K., & Hirasawa, N. (2021). Augmented reality-assisted femoral bone resection in total knee arthroplasty. JBJS Open Access, 6(3), e21-00001.
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Rath, B., Huber, T., Kamper, M., Beckmann, J., Hirschmann, M. T., Hammer, N., & Ullmann, D. (2026). Feasibility and accuracy of intraoperative 3D surface scanning to assess femoral component alignment relative to the native distal femoral articular surface in total knee arthroplasty. Knee Surgery, Sports Traumatology, Arthroscopy.
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World Model for Robotic Ultrasound Navigation

- **Mentors:** Yuan Bi and Xuesong Li
- **Topic Area:** Robotic ultrasound World models
- **Problem:** Robots move the probe without anticipating how the anatomy will change.
- **Background & Motivation:** Our SonoGraph-WM plans by imagining future scene graphs. Label maps keep richer anatomy.
- **Goal:** Extend SonoGraph-WM to predict future label maps and probe poses.
- **Proposed Solution:** Train on CT-derived label-map–pose sequences; imagine candidate paths and pick the shortest one reaching the goal.
- **Contact:** yuan.bi@tum.de & xuesong.li@tum.de
- **Tasks and Objectives**
  - 1. Generate label-map–pose data fromCT
  - 2. Train the label-map world model
  - 3. Plan and navigate in simulation
  - 4. Bonus: robot–phantom test
- **5 Related Works**
  - 1. Li et al., SonoGraph-WM, arXiv 2026
  - 2. Li et al., US scene graphs, MICCAI 2025
  - 3. Fan et al., US world model, arXiv 2026
  - 4. Yue et al., EchoWorld, CVPR 2025
  - 5. Hafner et al., PlaNet, ICML 2019

Teaser Image



SonoGraph-WM; this project replaces scene graphs with label maps.

Human-Centered Computing and Extended Reality

“Towards imitation learning for exoskeletons” - Developing a knee angle forecast engine





Mentors: Victor Schaack, Daniel Homm, Christina Valle

Topic Area: Movement analysis, prediction, robotic simulation

Background and Motivation: Increasing usability for exoskeletons requires the system to predict the wearers intent. This can be done by imitating normal human motion.

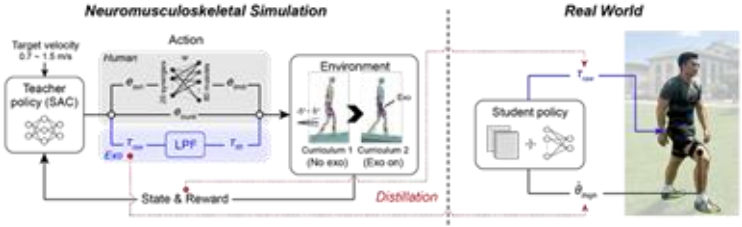
Problem: Current models are highly susceptible to different sensor placements and really heavily on EMGs. This increases intra-subject variability to too high a degree to inform a robotic control policy.

Goal: Develop a pipeline, in Isaac Lab or Mujoco, that predict knee angles using 2/4 arbitrarily placed IMUs per leg segment.

Proposed Solution: Teacher student policy, with the teacher policy having access to EMG data, optical tracking data and IMU data, while the student policy is distilled to down to only IMU data, just like deployed on the exoskeleton.

Tasks and Objectives:

1. (Confer with medical experts which movements are most relevant for rehabilitation use-case).
2. Dataset suitability (rehab-movement included, sensors, frequencies)
3. Synthetic data augmentation to suit exoskeleton IMU placement
4. Teacher-student distillation (EMG + optical tracking + IMU → IMU only)
5. ROS bridge that communicates knee angles for later robotic use.



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Human-Centered Computing and Extended Reality